PROO/SEW 3 - Assignment: *Billa Plus Plus*

Objective

Create a program for managing a supermarket's articles.

Things To Learn

- Working with *hash maps*.
- Working with *enums*.

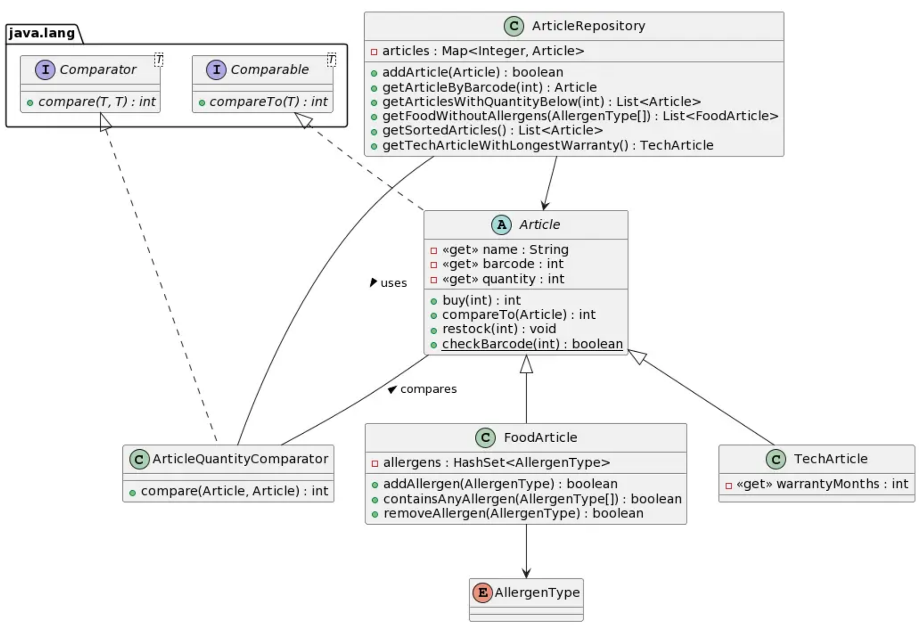
Submission Guidelines

- Your implemented solution as ***zipped*** *IntelliJ*-project.

Task

You are tasked with implementing the article management for the new supermarket *Billa Plus Plus* (or *B++* for short). Right now, the product range solely consists of *tech gadgets* and food items - but we need to keep in mind, that the assortment will surely grow in the near future!

Take a look at the following diagram and the implementation details below:



1. `Article`
   - The `abstract` base class for all articles the market sells.
   - Negative article quantities should be corrected to `0`.

- The quantity can be changed using the `buy` and `restock` methods, both throwing an `IllegalArgumentException` if a negative value is passed. Additionally, the `buy` method should return the quantity that was *actually* removed from stock.
- Implements the `Comparable`-interface using the article name for ordering.
- Provides a `static`-method for checking the validity of a barcode. See section below for details.
- Invalid barcodes passed to the constructor should lead to an `IllegalArgumentException`.
- A simple `toString`-method should provide a neat *String*-representation of an article.

2. `FoodArticle`
   - The class for all groceries, inheriting from `Article`.
   - Additionally stores the food's allergens in a `HashSet` using the ***enum**erated type* `AllergenType`.
     - Currently there are 14 food ingredients that must be declared as allergens: `A`, `B`, `C`, `D`, `E`, `F`, `G`, `H`, `L`, `M`, `N`, `O`, `P`, `R`.
   - The constructor accepts an array of allergens. The `HashSet`'s `addAll`-method could prove to be useful for this.
   - The class provides some methods for managing a food item's allergens. Both `addAllergen` and `removeAllergen` should return the *success* of the respective operation - an allergen can't be added twice, a non-existing allergen can't be removed - while `containsAnyAllergen` should return `true` if the food item has *any* of the passed allergens.
3. `TechArticle`
   - The class for all *tech gadgets*, inherits from `Article` and extends its properties by *warranty months*.
   - By law, *tech gadgets* are required to provide half a year of warranty. Invalid values should automatically be corrected.
4. `ArticleRepository`
   - Stores the `Article`s in a *hash map* using their barcodes as keys.
   - Provides various methods for adding and accessing articles:
     - `addArticle` adds an article to the map and returns `true` if successful. Articles need to have a unique barcode and a *positive* quantity!
     - `getArticleByBarcode` returns an article by the barcode passed.
     - `getArticlesWithQuantityBelow` returns a list of articles with the stock below a passed value. The articles should be sorted by their quantity. This can be achieved by implementing the `ArticleQuantityComparator`.
     - `getFoodWithoutAllergens` returns a list of articles (sorted by their names) not containing **either** of the passed allergens.
     - `getSortedArticles` returns a list of articles sorted by their names.
     - `getTechArticleWithLongestWarranty` returns the gadget with

the longest warranty. Returns `null` if there are no `TechArticle`s in the store.

EAN-8 Validation

The *EAN-8* barcode is used for small packages such as chewing gums or cigarettes - or in our case small data types like integers. This globally unique item number shares the same validation algorithm as its *big brothers* having 12, 13 and 14 digits.

The following diagram demonstrates the algorithm for the barcode *4534232X*:

Number	4	5	3	4	2	3	2	
Weight	1	3	1	3	1	3	1	
Products	4	15	3	12	2	9	2	
Sum								47
Sum *mod* 10								7
Difference to next 10								**3**

Thus, the complete valid barcode is *45342323*.